EXP 6

Create an R List Containing Vectors, Matrices, and Functions

Aim:

To create an R list containing vectors, matrices, and functions and display its contents.

Procedure:

1. Open RStudio or R console.
2. Create a numeric vector.
3. Create a matrix.
4. Define a function.
5. Store the vector, matrix, and function in a list.
6. Display the list contents.

Code:

```r
numbers <- c(10, 20, 30, 40)

matrix_data <- matrix(1:6, nrow = 2, ncol = 3)

square <- function(x) {
  return(x * x)
}

my_list <- list(
  numbers = numbers,
  matrix_data = matrix_data,
  square = square
)

print(my_list)
print(my_list$square(5))
```

Output:

```
$numbers
[1] 10 20 30 40

$matrix_data
     [,1] [,2] [,3]
[1,]    1    3    5
[2,]    2    4    6

$square
function (x)
{
    return(x * x)
}

[1] 25
```